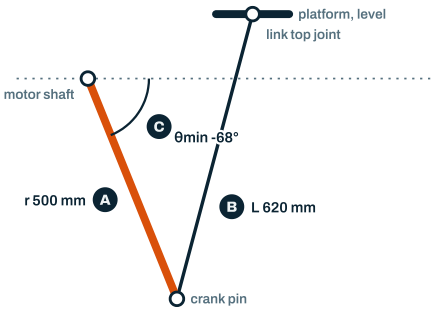
What to measure on the platform

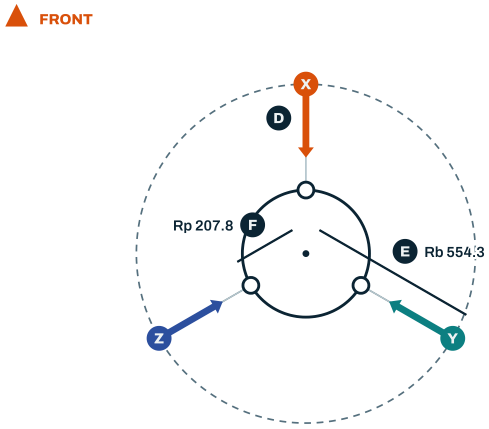
Seven numbers turn the millimetres and degrees from estimates into real values. It takes about fifteen minutes with a ruler, a phone angle app and the platform at its lowest level position. The drawings redraw from what you type, so you can check them against the real thing.

Before you measure: put the platform at its level minimum

1. Run the program, press `1` for Setup and let it home.
2. Press `1` again, jog each leg until the platform is level and as low as it goes, then press `n` to mark all three.
3. Stay there and measure. Do not touch the legs while measuring. Details in the run guide.



Side view of one leg, platform at its level minimum. Drawn from your numbers. The dotted line is the horizontal through the motor shaft.



Top view, looking down, front at the top. Solid circle: the platform, with a ring at each link's top joint. Dashed circle: the motor shafts (hidden when the two radii match). Coloured arrows: where each crank points when it is horizontal. The firmware is set for leg X at the front, Z at the back left and Y at the back right.

A Crank length r

From the centre of the motor shaft to the centre of the pin joint at the end of the crank arm. Use calipers or a steel rule; half a millimetre is plenty.

_____ mm

B Link length L

From the centre of one ball joint to the centre of the other, along the link. Not the length of the rod itself.

_____ mm

C Crank angle at the minimum θmin

With the platform at its level minimum, the angle of the crank arm (shaft to pin) against the horizontal. Use a phone angle app held along the arm. **Negative if the arm points below the horizontal**, positive if above. Check the other two legs read the same.

_____ deg

D Which way the crank points

Seen from above with the crank arm horizontal: pointing away from the centre line (outward), toward it (inward), or sideways. The top view shows the arrow.

Circle one: Outward / Inward / Sideways anticlockwise / Sideways clockwise

E Base radius

Horizontal distance from the middle of the platform to each motor shaft. Easier: measure the straight distance between two motor shafts and the radius is that divided by 1.732 (equal triangle).

_____ mm

Shaft to shaft: _____ mm

F Platform radius

Horizontal distance from the middle of the platform to each link's top joint, in the same way. Or measure between two top joints and divide by 1.732.

_____ mm

Joint to joint: _____ mm

G Which way more steps turn the crank

In Setup (minimums) select a leg and press `d` once: the step count goes up. Watch the crank pin. Does it rise or fall?

Circle one: Pin rises / Pin falls

Check the leg order

Stand where the front of the platform is. Which driver is wired to which leg? The firmware is set for the front leg X, the back-left leg Z and the back-right leg Y. Angles run anticlockwise from the front, seen from above. If your legs are wired differently, change these.

X leg at (deg)	Y leg at (deg)	Z leg at (deg)
0	240	120

One question, not a measurement

What stops the platform sliding sideways or turning? A centre pole or guide, a pivot, or only the three links? The model assumes the middle of the platform stays on the vertical axis. Write the answer here and bring it with the numbers.